

Data Characteristics

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HES 3000-002, Sport Performance Analytics

Understanding the Dataset

Start With the Dataset, Not the Numbers

- What is being measured?
- When and where were the data collected?
- What does one row represent?
- What does each column represent?

What Does One Row Represent?

One row could represent:

- One athlete
- One athlete during one season
- One athlete during one game
- One shot, pass, or possession
- One sensor measurement

The meaning of a row is the dataset's **unit of observation**.

The Same Sport, Different Units of Observation

Dataset	One row represents
Roster	One athlete
Season statistics	One athlete-season
Box score	One athlete-game
Shot chart	One shot attempt
Training load	One athlete-session
10 Hz GPS data	One athlete at 1/10 th of a second

Data Granularity

Granularity describes the level of detail in a dataset.

- Season totals provide a broad summary.
- Game-level data show changes across games.
- Event-level data describe individual actions.
- Sensor data may contain several measurements per second.

More detailed data provide more possibilities, but also require more storage, cleaning, and interpretation.

What Identifies Each Observation?

An identifier distinguishes one observation from another.

- Examples:
 - athlete_id
 - game_id
 - date
 - trial_number

An athlete's name may not be a reliable identifier:

- Names may be misspelled.
- Two athletes may share the same name.

Variables and Data Formats

Qualitative and Quantitative Data

Categorical variables

- Describe groups, labels, or states.
 - Position
 - Team
 - Play type
 - Availability status

Numerical variables

- Represent counts or measurements.
 - Minutes played
 - Jump height
 - Distance traveled
 - Number of shots

Categories

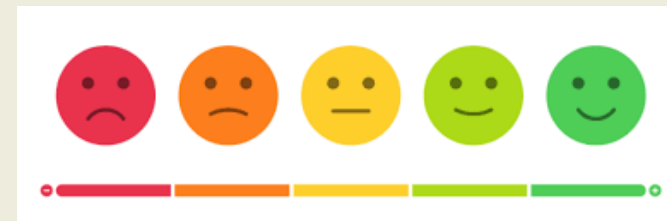
Nominal

- Categories without an inherent order:
 - Position
 - Team
 - Shot outcome
 - Injury type



Ordinal

- Categories with a meaningful order:
 - Low, moderate, high
 - First, second, third
 - Availability classifications
 - Subjective rating scales



Counts and Measurements

Discrete data

- Usually represent counts:
 - Rebounds
 - Goals
 - Sprint efforts
 - Shot attempts



Continuous data

- Usually represent measurements:
 - Speed
 - Distance
 - Time
 - Force
 - Jump height



Dates and Times

Dates and times may represent:

- Competition date
- Collection date
- Session start time

Common problems include:

- 09/10/26 versus 10/09/26
- Dates stored as text
- Inconsistent time zones

Units and Definitions

A number has little meaning without a definition and unit.

- The unit can completely change what is being measured
 - Is speed measured in mph, km/h, or m/s?
 - Is body mass recorded in pounds or kilograms?
 - Is duration recorded in seconds or minutes?

Data Dictionary

- A data dictionary explains the variables in a dataset.

Variable	Meaning	Unit or values
athlete_id	Unique athlete identifier	Text
duration	Time spent participating	Minutes
distance	Total tracked distance	Meters
session_rpe	Athlete-reported exertion	1–10
player_load	Total tracked workload	AU

Data Structure

Wide and Long Data

Wide format

- Multiple measurements appear in separate columns.

Athlete | Week_1 | Week_2 | Week_3

Long format

- Each measurement appears in a separate row.

Athlete | Week | Jump_Height

Wide vs Long :

- Wide data are often easier to read in a spreadsheet.
- Long data are usually easier to group, visualize, and analyze.

Wide vs Long?

- Which would be better for:
 - Plotting jump height across ten weeks → Long
 - Comparing the most recent test for every athlete → Either
 - Entering scores for a small number of assessments → Wide may be convenient
 - Adding new testing dates throughout a season → Long

Raw Data and Summary Data

Raw data

- Individual events or measurements
- Greater detail
- More opportunities for checking and reanalysis
- Eg: every jump trial

Summary data

- Totals, averages, or maximums
- Easier to interpret quickly
- May hide variation or unusual observations
- Eg: each athlete's average or maximum jump height

One Dataset, Multiple Tables?

- Sport data may be stored across several related tables:
 - Athlete information
 - Games or training sessions
 - Performance test results
- Trying to place everything in one spreadsheet can produce:
 - Repeated athlete information
 - Inconsistent values
 - Excessively wide datasets
 - Confusion about what each row represents

Common File Formats

- CSV
 - Plain-text table
 - Easy to import into most software
 - Does not preserve formatting, formulas, or multiple sheets
- Excel
 - Supports multiple sheets, formulas, and formatting
 - Convenient for manual data entry
 - Makes it easier to create structures that look good but analyze poorly

Common Problems

Spreadsheet Designed for People

A spreadsheet may be readable to a person but difficult to analyze.

- Common examples:
 - Merged cells
 - Multiple header rows
 - Blank rows separating groups
 - Totals mixed with individual observations

Statistical software cannot interpret meaning that exists through visual formatting.

One Column, One Type of Information

- Problematic cells might contain:
 - 185 lb
 - Injured - ankle
 - DNP
 - Smith, John #24
- These combine values, units, notes, or identifiers.
- One column should have one consistent meaning

name	jersey_number	weight_lbs	status	injury_location
John Smith	24	185	Injured	ankle

Inconsistent Categories

- The same category might appear as:
 - Guard
 - guard
 - G
 - G (with an extra space)
 - Point Guard
- To a person, these may appear equivalent but to software, they are different categories.
- Categories need consistent spelling, capitalization, and definitions.

Other common data problems

- Duplicate observations: truly identical or represent separate events?
- Impossible and suspicious values
- Missing data: truly missing or something else?

Data Hygiene

Basic Data Hygiene

Data hygiene means keeping data organized, consistent, and understandable.

- Before analysis:
 1. Preserve the original file.
 2. Create a separate working copy (in excel, copy sheet).
 3. Identify the meaning of each row and column. (data dictionary)
 4. Standardize categories, formats, and units.
 5. Check duplicates, ranges, and missing values.
 6. Document important changes.

Summary & Looking Ahead

- Understand the dataset before calculating anything.
 - Structure determines what questions the data can answer.
 - Values require definitions, units, and context.
 - Missing values can represent different situations.
 - Data inspection should occur before analysis.
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- Next: data visualization